

Physics 7501 Quiz 3
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Sharon Ye

(2.3.1) Spin Precession

Under a magnetic field along the z -axis, described by a Hamiltonian

$$\hat{H} = -B\hat{Z},$$

where

$$\hat{Z} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

is the 3rd Pauli matrix. Consider an initial state whose spin magnetic moment is along the x -axis. What is the spin precession periodicity for its magnetic moment? Express your answer as a function of B . You can set $\hbar = 1$.

Recall that the magnetic dipole moment $\vec{\mu}$ is proportional to the spin angular momentum $\vec{\sigma}$ (point in the same direction). Our initial state points along the x -axis (without loss of generality assume $+x$ -direction) thus

$$|\psi(0)\rangle = |+x\rangle = \frac{1}{\sqrt{2}} (|+z\rangle + |-z\rangle)$$

$\Rightarrow$

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}} \left(e^{iBt} | +z \rangle + e^{-iBt} | -z \rangle \right)$$

We want to find T such that $|\psi(T)\rangle = |\psi(0)\rangle$, thus

$$\frac{1}{\sqrt{2}} \left(e^{iBT} | +z \rangle + e^{-iBT} | -z \rangle \right) = \frac{1}{\sqrt{2}} (| +z \rangle + | -z \rangle)$$

$\Rightarrow$

$$\frac{1}{\sqrt{2}} \left(| +z \rangle + e^{-2iBT} | -z \rangle \right) = \frac{1}{\sqrt{2}} (| +z \rangle + | -z \rangle)$$

$\Rightarrow e^{-2iBT} = 1 \Rightarrow 2BT = 2\pi \Rightarrow T = \frac{\pi}{B}$. Since time should be positive, we take the absolute value

$$T = \frac{\pi}{|B|}$$

(2.3.2) Neutrino oscillation

Both neutrino oscillation and spin precession are described by 2-level Hamiltonians of the same form. Consider two “mass eigenstates” $|\nu_1\rangle$ and $|\nu_2\rangle$ in neutrino oscillation: do they correspond to

Pauli- $\hat{Z}$ eigenstates or Pauli- $\hat{X}$ eigenstates in the spin precession problem? Explain.

$|\nu_1\rangle$ and $|\nu_2\rangle$ are analogous to the Pauli- $\hat{Z}$ eigenstates since they are energy eigenstates of their associated Hamiltonian.

(2.3.3) 2-qubit Hamiltonian

Find the smallest eigenvalue of the 2-qubit Hamiltonian

$$\hat{H} = \hat{Z}_1 \otimes \hat{Z}_2.$$

For any two operators $\hat{A}$ and $\hat{B}$, if λ_a is an eigenvalue of $\hat{A}$ and λ_b is an eigenvalue of $\hat{B}$ i.e. $\hat{A}|a\rangle = \lambda_a|a\rangle$ and $\hat{B}|b\rangle = \lambda_b|b\rangle$, then

$$(\hat{A} \otimes \hat{B})(|a\rangle \otimes |b\rangle) = \hat{A}|a\rangle \otimes \hat{B}|b\rangle = \lambda_a \lambda_b (|a\rangle \otimes |b\rangle)$$

$\implies \lambda_a \lambda_b$ is an eigenvalue of $\hat{A} \otimes \hat{B}$ (pairwise products of the eigenvalues of $\hat{A}$ and $\hat{B}$). Since the eigenvalues of $\hat{Z}$ are ± 1 , the eigenvalues of $\hat{H}$ are also $\pm 1 \implies \lambda_{\min} = -1$